

couplr: Optimal Matching with Verifiable Certificates and Sparse Edge Generation

by Gilles Colling

Abstract Matching two groups of units so that the total cost of the chosen pairs is as small as possible is the linear assignment problem, which sits underneath causal-inference matching and experimental design. Two things bound what software built on it can offer. An answer arrives without a proof: a solver reports the state it terminated in, not a checkable statement about the matching. And the graph of admissible pairs is built before it is solved, so its memory sets the size ceiling. We present the couplr package, which addresses both. Every flow-representable matching design compiles into one internal flow model, so node potentials come back from all of them and can be checked against the linear-programming optimality conditions. On those potentials the package optimises over a restricted set of pairs: each unit starts with its nearest admissible partners, the omitted pairs are priced against the duals the sparse solve returns, and pairs enter until none prices in, at which point the duals certify the result optimal for the complete problem. Matching 16,667 treated units to 33,333 controls holds 0.29 percent of the pairs and returns a certified optimum. Warm starting the same loop traces a matching frontier over a swept constraint at roughly half the cost of solving each point cold. A data-frame interface, nineteen assignment algorithms, balance diagnostics and cardinality matching share the same core. couplr is available on CRAN.

1 Introduction

Many data-analysis tasks come down to the same question: given two sets of objects and a cost attached to every possible pair, which pairing has the smallest total cost? A researcher building an observational study pairs treated units with controls that resemble them on measured covariates, a scheduler assigns staff to shifts, an ecologist pairs surveyed plots across two time points, and an image-processing pipeline maps the pixels of one picture onto the pixels of another. In each case the mathematical object is the linear assignment problem, and the quality of the answer depends on solving it to optimality rather than greedily.

R has good software at both ends of this problem, in two separate groups of packages. On the statistical side, **MatchIt** (Ho et al., 2011) provides a common interface to several matching designs and delegates optimal pair and full matching to **optmatch** (Hansen, 2007), which formulates those designs as minimum-cost flow problems (Hansen and Klopfer, 2006), reaches RELAX-IV or one of four LEMON algorithms through its solver argument, and stores sparse distances as an `InfinitySparseMatrix` whose non-finite entries are the pairings it will not consider. **Matching** (Sekhon, 2011) provides multi-variate and propensity-score matching, with `GenMatch()` using genetic search to optimise covariate weights. **designmatch** (Zubizarreta et al., 2024) formulates balance-constrained designs as integer programs. **rcbalance** (Pimentel, 2022) matches under refined covariate balance constraints (Pimentel et al., 2015), taking a distance structure that lists the permissible controls of each treated unit and solving the resulting network as a minimum-cost flow problem. **quickmatch** (Sävje et al., 2024) aims at large data sets with generalized full matching, which its own description calls near-optimal. **cobalt** (Greifer, 2025) supplies balance diagnostics across packages. What these expose is the matching design; the assignment problem underneath is reached through a solver argument at most, and alternative optima and a bottleneck objective sit outside all of them. Statements about other packages here and in Table 6 were assessed on 2026-08-09 against **MatchIt** 4.7.2, **optmatch** 0.10.8, **Matching** 4.10-15 and **designmatch** 0.5.5, and against the reference manuals of **rcbalance** 1.8.8 and **quickmatch** 0.2.3 on 2026-08-26.

On the algorithmic side, **clue** (Hornik, 2024) exposes the Hungarian method through `solve_LSAP()`, and **lpSolve** (Berkelaar et al., 2024) solves the assignment problem as a general linear program. These return an optimum for a cost matrix the user has already built; covariate handling, calipers and balance tables sit outside their scope.

A workflow that needs both halves therefore threads several packages by hand. Matching on observed covariates instead of a propensity score, imposing calipers on more than one variable at a time, blocking, and choosing the solver deliberately are all available in R, each in a different package.

Two further limits are common to both groups, and they are the ones this article is about. The first concerns what a result says about itself. A solve returns a matching and a status, and the status records the state the solver terminated in. Whether that matching is the optimal one is a separate question with a checkable answer, since linear programming supplies optimality conditions a third party can test on the returned solution, and none of the packages above exports a function returning the dual

variables those conditions need or checking the conditions. `optmatch` comes closest: a `fullmatch()` result carries the node prices its flow solver ended on in an undocumented attribute, and the function that would evaluate the primal against them is internal. What it documents in their place is a solver tolerance, `tol` times the number of subjects bounding how far the returned match may differ from an optimal solution.

The second concerns size. Sparse matching is a developed literature, and what its methods have in common is that the sparse network is settled before the matching is solved over it. `rcbalance` takes a per-unit list of permissible controls built from exact groups and a caliper. Yu et al. (2020) choose a propensity-score caliper by an iterative form of Glover's algorithm and match optimally in the much sparser graph that leaves, and Yu and Rosenbaum (2022) grade the potential pairings and admit lower grades only as far as they are needed. In each case the rule deciding which pairs exist is fixed in advance of the solve that uses them, and a network too sparse drops pairings it should have considered. What is missing is a rule read off the solution in hand, and a statement about the pairs it never built.

We present the `couplr` package (Colling, 2026), which treats the assignment problem itself as the organising object and builds the workflow around it. A user who starts from a data frame with covariates and a treatment indicator, and a user who starts from a cost matrix, enter through different functions and reach the same solver core. We contribute five things to R:

1. A common compilation layer. One-to-one, k-to-one, with replacement, blocked and full matching are the same internal flow model under different arc capacities, so what is written once serves all of them.
2. Certificates rather than reported status. `verify_assignment()` and `verify_flow()` check primal feasibility, dual feasibility, complementary slackness and objective equality on a solution, reporting each separately and independently of the solver that produced it, in exact arithmetic where the potentials allow and naming the arithmetic the verdict was reached in.
3. Dual-certified adaptive edge generation. Under `memory_mode = "implicit"` the complete graph is never built: a sparse restricted problem is solved, the omitted pairs are priced against its duals, violating pairs are added, and the loop repeats until none prices in, at which point the duals certify optimality for the complete problem.
4. Warm-started matching frontiers. `match_path()` sweeps one constraint over a sequence of values, resuming each point from the one before it, with a certificate per point.
5. Nineteen assignment algorithms written from scratch in C++ and reachable by name, covering augmenting-path, auction, cost-scaling and network-flow families, with k-best and bottleneck assignment, entropy-regularised transport, and cardinality matching that reports an optimality gap against the largest sample its constraints admit.

The article is arranged for two kinds of reader. The assignment engine and the matching interface built on it come first and stand on their own; the optimisation architecture, its implementation and the measurements follow for a reader interested in where the certificates and the sparse edge generation come from.

Results were produced with `couplr` version 1.7.0.

2 The core assignment engine

The assignment problem

Let $C \in \mathbb{R}^{n \times m}$ be a cost matrix whose entry C_{ij} is the cost of pairing row $i \in \{1, \dots, n\}$ with column $j \in \{1, \dots, m\}$, and take $n \leq m$; `couplr` transposes the problem internally when it is not and maps the assignment back afterwards, so the user never sees the transpose. Write $X \in \{0, 1\}^{n \times m}$ for the assignment matrix, with $X_{ij} = 1$ when row i is matched to column j . The linear assignment problem is

$$\min_X \sum_{i=1}^n \sum_{j=1}^m C_{ij} X_{ij} \quad \text{subject to} \quad \sum_j X_{ij} = 1 \quad \forall i, \quad \sum_i X_{ij} \leq 1 \quad \forall j. \quad (1)$$

Every row is matched and every column is used at most once. The row constraint is an equality: relaxing it to $\sum_j X_{ij} \leq 1$ would leave $X = 0$ optimal for any non-negative cost matrix, which is not the problem a matching application poses.

Forbidden pairs are encoded as $C_{ij} = \infty$, and enough of them make (1) infeasible, since no assignment can then reach every row. `couplr` solves a lexicographic relaxation in that case: the largest number of matched rows first, then the smallest total cost among the assignments attaining that number. The section on constraints below describes how that problem is reached and what it costs.

The constraint matrix of (1) is totally unimodular, so the linear programming relaxation has integral optima (Birkhoff, 1946; Burkard and Çela, 1999) and the combinatorial problem can be solved by primal-dual methods rather than by branch and bound. The dual assigns a potential u_i to each row and v_j to each column, and is

$$\max_{u,v} \sum_{i=1}^n u_i + \sum_{j=1}^m v_j \quad \text{subject to} \quad u_i + v_j \leq C_{ij} \quad \forall (i,j), \quad v_j \leq 0 \quad \forall j. \quad (2)$$

The sign condition is what the column inequality in (1) costs: the dual objective sums v over every column while an assignment pays only for the ones it uses. When $n = m$ none can be left out, the column constraints bind, and v is unrestricted in sign; `verify_assignment()` imposes $v_j \leq 0$ only when $n < m$, since Jonker-Volgenant returns free-sign potentials on a square problem and they certify optimality there.

Complementary slackness ties a pair of potentials to a particular X :

$$X_{ij} = 1 \implies u_i + v_j = C_{ij}, \quad \sum_i X_{ij} = 0 \implies v_j = 0. \quad (3)$$

The reduced cost $C_{ij} - u_i - v_j$ is a lower bound on what forcing pair (i, j) would add to the total rather than the whole of it, because forcing one pair can displace others.

Verifiable optimality certificates

Dual feasibility on its own says nothing about a particular X . What certifies optimality is X primal feasible, (u, v) feasible for (2), and both halves of (3) holding; in exact arithmetic those three settle the objectives as well, so objective equality is a reported check rather than a condition of its own. `verify_assignment()` reports the four separately, so a failure names the check it failed on.

```
set.seed(2)
cost <- matrix(runif(120), nrow = 6, ncol = 20)
verify_assignment(assignment(cost), cost)

#> Assignment certificate
#> =====
#>
#> primal_feasible      TRUE
#> dual_feasible        TRUE
#> complementary_slackness TRUE
#>   matched arcs tight  TRUE   (max slack 0.000e+00)
#> unmatched columns free TRUE   (max |v_j| 0.000e+00)
#> duality_gap          0.000000e+00
#> certified_optimal    TRUE
#> arithmetic           exact, no tolerance
#>
#> primal objective    0.2821665586
#> dual objective      0.2821665586
#> matched            6 of 6 rows, 20 columns
```

Two properties make this a check rather than a report. It is written against the solution rather than the solver, taking an assignment, a cost matrix and a pair of potentials and never consulting the code that produced them. And optimal duals are shared by all optimal solutions of a linear program, so a matching from one solver can be certified against potentials from another, which is what puts the solvers returning no duals of their own inside its scope.

Complementary slackness is where a plausible verifier goes wrong, and the rectangular case is where it shows. Asserting the first half of (3) alone, tightness on matched pairs, accepts solutions that are not optimal: in our own prototyping a perfect, dual-feasible, matched-tight solution passed such a check with its primal cost 0.4% above the optimum, because one unmatched column retained $v_j < 0$ and contributed a spurious term to the dual objective. `verify_assignment()` asserts both halves and reports them as `cs_matched_tight` and `cs_unmatched_free`.

The arithmetic the conditions are decided in is the other place where a certificate can overstate what it knows. Each of them is the sign of $C_{ij} - u_i - v_j$, and every quantity in that expression is a double, which is a rational number, so each sign has an exact answer. Deciding them within a band around zero instead makes the verdict a statement about a neighbourhood of the instance rather

than about the instance, and what the band costs can be written down. A certificate at tolerance ϵ accepts potentials that violate dual feasibility by ϵ on any pair, miss tightness by ϵ on any matched arc, and leave $|v_j| \leq \epsilon$ on any unused column. Shifting to $u_i - \epsilon$ and $v_j - \epsilon$ restores exact dual feasibility and the sign condition while costing the dual objective $(n + m)\epsilon$, so the optimum is at least $\sum_i u_i + \sum_j v_j - (n + m)\epsilon$. Tightness within ϵ on the n matched arcs and $|v_j| \leq \epsilon$ on the $m - n$ unused columns put the matching's own cost at most $m\epsilon$ above that same sum. Its excess over the optimum is therefore at most $(n + 2m)\epsilon$. At the largest shape in the benchmark below and the default $\epsilon = 10^{-9}$, that is 8×10^{-5} in the cost units.

Setting the band to zero is not the fix, because the sign of the double evaluation can itself be wrong. With $C_{ij} = \pm 2^{-60}$, $u_i = 1$ and $v_j = -1$ the first subtraction rounds to -1 and the expression returns zero on a pair whose reduced cost is not, which in the negative case is a dual infeasibility a zero-band check would accept. `verify_assignment()` evaluates the sign exactly instead: it splits each rounded addition into the sum and the part rounding lost, giving an expansion of doubles equal to the difference in the reals, and takes the sign of that expansion (Shewchuk, 1997). A rounding-error bound decides which pairs the double evaluation already settles, so the expansion runs near tightness rather than on all nm .

Removing the band from the two conditions removes it from the conclusion. Objective equality is then not a fourth measurement but a consequence: with matched arcs tight and unmatched columns free, the primal cost is the sum of u over matched rows plus the sum of v over used columns, which is the dual objective exactly when every row is matched. The software reports the gap it computes as an independent numerical cross-check on that reasoning.

An exact certificate states that the matching attains the minimum total cost of the matrix as supplied, and is available when the potentials are exactly optimal for it, which is a property of the arithmetic that produced them. Over ten instances per cell, integer costs and uniform costs on the unit interval gave one in every instance at 50, 200 and 800 rows square, at 200 rows against 600 columns, and for each of the eleven solvers we ran at 200 by 200. Euclidean distances between normal clouds gave one in none: those potentials miss exact tightness by a relative 2×10^{-16} to 5×10^{-15} , which is what rounding in the arithmetic that produced them costs, and there the certificate decides the same conditions within a stated tolerance and says so. Every instance was certified in one arithmetic or the other, and a verdict names which; `arithmetic = "exact"` refuses the fallback for a caller who wants the stronger statement or nothing.

The nineteen-solver portfolio

Every solver of (1) returns the same optimal value, so the choice among them is a performance question rather than a statistical one, and it depends on the shape of the problem. We group them into four families and a set of special-purpose routines.

Augmenting-path methods grow a matching one row at a time, each time finding a shortest augmenting path in the residual graph under reduced costs. The Hungarian method (Kuhn, 1955; Munkres, 1957) is the classical member; the Jonker-Volgenant algorithm (Jonker and Volgenant, 1987) adds column reduction and auction-style initialisation, which is what makes it fast in practice on dense problems despite an $O(n^3)$ worst case. Lapmod is the sparse variant, working from an adjacency list so that forbidden pairs cost nothing to carry.

Auction methods (Bertsekas, 1988) let rows bid for columns, raising prices until no row wants to switch. They converge to optimality once the bid increment ϵ falls below $1/n$ for integer costs, and ϵ -scaling gives the useful practical variants. Auction is naturally parallel and tolerant of near-degenerate cost structures where augmenting-path methods spend time on long paths.

Cost-scaling methods solve a sequence of increasingly precise problems, each time halving the scale on which costs are rounded and reusing the previous solution (Goldberg and Kennedy, 1995). The Gabow-Tarjan bit-scaling algorithm (Gabow and Tarjan, 1989) belongs here. On a graph with $|V|$ nodes and $|E|$ edges it reaches $O(\sqrt{|V|} |E| \log(|V| C_{\max}))$ for integer costs, where $C_{\max} = \max_{i,j} |C_{ij}|$. A complete bipartite graph on n and m units has $|V| = n + m$ and $|E| = nm$, so the bound there is $O(\sqrt{n+m} nm \log((n+m) C_{\max}))$. To our knowledge this algorithm did not previously have a publicly available open-source implementation in any language.

Bit-scaling runs on integers, so the conversion is part of the method and its rule is stated rather than left to the caller. Finite costs are shifted so the smallest is zero, multiplied by a scale factor and rounded, with the factor set to keep the scaled costs clear of the sentinel marking a forbidden pair and the path sums the solver forms inside 64-bit arithmetic. A matrix whose finite entries are each within 10^{-9} of an integer enters unscaled, at the integer nearest each entry. Where those entries are integers the conversion is exact and the optimum returned is the optimum of the instance as supplied; where they are merely near one, each cost moves by at most 10^{-9} , so the matching returned costs at

most $2 \min(n, m) \times 10^{-9}$ more than that optimum. A matrix whose range leaves no usable factor is refused with the limit named. Costs that are not integers are solved on the rounded instance, and the matching returned costs at most about 10^{-7} of the cost range more than the optimum of the matrix as supplied at $n = 1000$, and 10^{-5} of it at $n = 10,000$ on a square problem. Potentials are divided by the factor and shifted back, so they belong to the original matrix, and whether the matching is optimal for it rather than only for the rounded instance is a question `verify_assignment()` answers.

Network-flow methods treat (1) as a minimum-cost flow problem on a bipartite network (Ahuja et al., 1993) and apply push-relabel (Goldberg and Tarjan, 1988), cycle cancelling, network simplex, or Orlin's strongly polynomial algorithm (Orlin, 1993). These are the most general formulations, which makes them the right base for variable-ratio designs where a group absorbs several units. On a plain one-to-one problem they are the slowest choice.

Special-purpose routines cover cases where the general machinery is wasted: exhaustive enumeration below nine units, Hopcroft-Karp bipartite matching (Hopcroft and Karp, 1973) when costs are binary or constant, Ramshaw-Tarjan (Ramshaw and Tarjan, 2012) for strongly rectangular problems where padding to a square matrix would dominate the work, and bottleneck assignment, which minimises the largest pair cost instead of the sum.

The binary case is worth stating in full, since a maximum-cardinality routine carries no costs and the problem does. When every admissible edge holds the same value, all perfect matchings cost the same and any maximum-cardinality matching is optimal. When the finite costs are $\{0, 1\}$, the routine runs Hopcroft-Karp on the zero-cost edges alone: a perfect matching there totals zero, which no assignment improves on. If that subgraph admits no perfect matching the optimum needs at least one unit edge, and the problem passes to the weighted solver on the original matrix. The specialised path answers only where its answer is the optimum.

Two extensions return something other than a single optimal assignment. Murty's ranking procedure (Murty, 1968) with Lawler's partition scheme (Lawler, 1972) enumerates the k cheapest assignments, which lets a user see how much of the total cost is forced and how much is arbitrary. Sinkhorn iteration (Cuturi, 2013) solves the entropy-regularised relaxation, returning a doubly stochastic coupling rather than a permutation.

Matrix-level interface and extensions

The solver layer is reachable on its own, for a user who arrives with a cost matrix. `lap_solve()` and `assignment()` take a matrix and return the optimal assignment, and `lap_solve()` also accepts a long data frame of source, target and cost columns, the natural shape when costs come from a database. The `method` argument names any of the nineteen solvers and defaults to "auto"; it accepts twenty names, since "ssp" is a second spelling of "sap".

```
set.seed(1)
cost <- matrix(sample.int(100, 25, replace = TRUE), nrow = 5)
res <- lap_solve(cost)
res

#> Assignment Result
#> =====
#>
#> # A tibble: 5 x 3
#>   source target cost
#>   <int>   <int> <int>
#> 1     1     4     7
#> 2     2     2    14
#> 3     3     1     1
#> 4     4     3    54
#> 5     5     5    44
#>
#> Total cost: 120
#> Method: bruteforce
```

Dual potentials come from `assignment_duals()`, which returns u and v alongside the assignment, so passing that result to `verify_assignment()` certifies the matching without a second solve. Four extensions operate on the same matrix: `lap_solve_kbest()` returns the k cheapest assignments in increasing order of cost, `bottleneck_assignment()` minimises the largest pair cost rather than the sum, `lap_solve_batch()` solves a stack of independent problems, optionally across threads, and

`sinkhorn()` returns the entropy-regularised coupling, which `sinkhorn_to_assignment()` rounds to a permutation.

```
kb <- lap_solve_kbest(cost, k = 3)
tapply(kb$total_cost, kb$rank, unique)
```

```
#> 1 2 3
#> 120 120 150
```

The two cheapest assignments both cost 120 and the third costs 150, so the optimum here is attained by two distinct pairings. A matching study reading that output learns its matched sample is not unique, which stays invisible when a solver returns one assignment. The solver portfolio and the dispatcher that selects within it are measured across a grid of cost regimes, admissibility patterns and aspect ratios in the empirical evaluation.

3 Matching with couplr

A matching workflow

The package ships `hospital_staff`, a synthetic personnel dataset supporting a treated/control workflow without external data: 200 units against 300, with age, experience, hourly rate, department, certification level and a full-time indicator.

`match_couples()` is the front door. It takes the two data frames and the covariates to match on and returns the optimal one-to-one pairing under the requested distance; `auto_scale = TRUE` screens the covariates for degenerate variance and excessive missingness and picks a scaling method, so that variables measured in years and in ordinal levels contribute comparably.

```
library(couplr)
data(hospital_staff)
treated <- transform(hospital_staff$nurses_extended, id = nurse_id)
control <- transform(hospital_staff$controls_extended, id = nurse_id)
covars <- c("age", "experience_years", "certification_level")

m <- match_couples(treated, control, vars = covars, auto_scale = TRUE)
m$pairs[1:3, c("left_id", "right_id", "distance", ".age_diff")]

#> # A tibble: 3 x 4
#>   left_id right_id distance .age_diff
#>   <chr>   <chr>     <dbl>     <dbl>
#> 1 1      260      0.217      -1
#> 2 2      294      0.175       2
#> 3 3      212       0         0
```

The call matches all 200 treated units at a total distance of 59.95. Beside the identifiers and each pair's distance, `m$pairs` carries a signed per-variable difference for every matching variable, so which covariate drives a poor pair is visible without recomputing anything; `m$unmatched` holds what is left over on each side and `m$info` records the method, the pair count, the total distance and the estimand.

`balance_diagnostics()` computes standardised mean differences, variance ratios and Kolmogorov-Smirnov statistics on the matched sample, and `balance_table()` formats them: experience and certification fall below the conventional 0.1 threshold on the absolute standardised difference (Stuart, 2010) and age sits slightly above at 0.116. `join_matched()` returns the analysis-ready frame, one row per pair, with every column of both inputs under `_left` and `_right` suffixes.

The matched frame is the input to an outcome analysis rather than the end of one. A paired difference taken on it estimates an adjusted association, and reading that as a causal effect is what rests on assumptions no matching method can verify: conditional ignorability, positivity, and the stable unit treatment value assumption. What matching can offer against the first is `sensitivity_analysis()`, which implements Rosenbaum bounds for the Wilcoxon signed-rank statistic on one-to-one matched pairs (Rosenbaum, 2002), reporting p-value bounds across a grid of the odds-ratio parameter Γ so that a reader can judge how strong an unmeasured confounder would have to be to overturn the result. The supplementary material carries both on the matched sample above.

Calipers, forbidden pairs and partial matching

Three mechanisms restrict which pairs are admissible, and all three work by writing non-finite entries into the cost matrix before it reaches the solver. A global `max_distance` forbids any pair beyond a distance ceiling. Per-variable calipers forbid pairs whose raw difference on a named variable exceeds a threshold, independently of the distance metric in use. Explicit forbidden pairs are passed as non-finite entries by the user. Every solver recognises a non-finite entry as a missing edge, so no returned pair ever violates a constraint.

Tight constraints usually make (1) infeasible. `couplr` first drops the rows and columns whose every edge is forbidden, reporting them unmatched. The remaining submatrix can still fail Hall's condition, with several rows competing for the same few columns and no assignment reaching every row. Rather than return whichever pairs a heuristic happens to find, `couplr` replaces the forbidden entries with a finite sentinel σ and re-solves with the same optimal solver.

Write $[\ell, h]$ for the range of the admissible costs of the pruned submatrix and k for the smaller of its two dimensions.

Lemma 1. *If $\sigma > k(|\ell| + |h|)$, then a padded assignment using more sentinel edges than another costs strictly more.* The proof is in the supplementary material.

A minimum-cost solve of the padded problem therefore uses as few sentinel edges as any assignment can, which is a maximum-cardinality admissible matching, and among the assignments attaining that number it minimises the real cost. Dropping the sentinel pairs leaves the lexicographic optimum: the largest admissible matching, cheapest among those of its size. `couplr` takes $\sigma = (k + 1)(|\ell| + |h|) + 1$, which satisfies the hypothesis in every sign case, and solves a maximisation by negation with the sentinel negated alongside it. No sentinel enters a reported cost.

Above a padded objective of 2^{53} `couplr` refuses the padding rather than returning an ordering its arithmetic may not support: `match_couples()` warns and falls back to a greedy partial matching, saying that it is not optimal, and `assignment(cardinality = "maximum")` errors and asks for rescaled costs or an explicit `unmatched_penalty`. That threshold is a precondition on the solve and not a proof of it. What establishes that a particular padded solve reached its optimum is the certificate, and the lemma carries that verdict across to the lexicographic objective; the supplementary material sets out both.

```
m_cal <- match_couples(treated, control, vars = covars, auto_scale = TRUE,
                      calipers = list(age = 3, experience_years = 2),
                      max_distance = 1.5)
```

The caliper costs pairs: 197 of the 200 treated units keep a partner. That is a caliper working, and the reason the unmatched identifiers are returned rather than dropped, since the units that fail to match define the population the estimate applies to. The constraints here are severe, leaving 2,173 admissible pairs out of 60,000, so the padded solve is what produces this result; a greedy pass over the same edges matches 180 units rather than 197.

Blocking is the other structural constraint. `block_id` restricts matching to levels of a factor, and `matchmaker()` builds blocks when no grouping variable exists, from an existing variable or by k-means or hierarchical clustering on chosen block variables. The solver runs once per block, and blocked designs report per-block balance so imbalance inside one stratum is not averaged away across the others.

Matching designs

Beyond one-to-one matching, `match_couples()` takes ratio for k-to-one designs and replace for matching with replacement. Five further designs have their own entry points, each returning an object in the same family. `ps_match()` estimates a propensity score (Rosenbaum and Rubin, 1983) and matches on it, with the caliper expressed in standard deviations of the linear predictor. `full_match()` assigns every unit to a matched group of variable size, so no unit is discarded, solving a minimum-cost flow problem with Johnson potentials (Johnson, 1977) on its optimal path and a greedy two-pass heuristic on large problems. `cem_match()` implements coarsened exact matching (Iacus et al., 2012), and `subclass_match()` divides the sample into propensity-score subclasses, returning subclass weights for the average treatment effect on the treated or on the whole sample.

`cardinality_match()` targets the cardinality-matching objective (Zubizarreta et al., 2014), the largest matched sample meeting prespecified balance constraints, with total distance ordered after cardinality; which solver answers it follows from the kind of balance asked for. Fine and refined covariate balance (Rosenbaum et al., 2007; Pimentel et al., 2015), stated through fine and refined, are representable in the network as category nodes, so such a call reaches a single minimum-cost

flow solve returning the largest balanced sample with a dual certificate at polynomial cost. Linear moment constraints, a finite `max_std_diff` or an explicit moments argument, are not, and are dualised and searched by branch and bound under `node_limit` and `time_limit`. Either way the result reports `n_matched` beside `best_possible`, the largest sample the constraints admit, the gap between them, whether the answer is certified, and what the search stopped on. A pruning loop is available as `engine = "heuristic"` and reports its gap as unknown, which is the honest reading of what a heuristic establishes.

Interoperability

`couplr` fits into a workflow built around the established packages. `as_matchit()` converts a `couplr` result into a `matchit` object and `match_data()` returns data in the layout `MatchIt` users expect, so downstream code keeps working; `bal.tab()` methods are registered for the `couplr` classes, so `cobalt` produces its usual balance output; and weighted matched data flows into `marginalEffects` (Arel-Bundock, 2025).

4 Common optimization architecture

Three of the package's capabilities are consequences of one decision. Compiling every flow-representable design into a single flow model is what makes node potentials available from all of them; the potentials are what certify a solution; a certificate is what makes it safe to optimise over a restricted set of pairs; and a certified restricted solve is what a warm start resumes from:

common flow duals $\rightarrow$ certificates $\rightarrow$ safe edge generation $\rightarrow$ warm-started paths.

Flow compilation

Underneath the designs of the previous section there is one object. It has nodes, each carrying a signed supply, and arcs, each carrying a cost and a capacity range. As Table 1 sets out, a one-to-one match, a k-to-one match, matching with replacement and a full match differ in the bounds their arcs receive, not in the code that solves them. Formulating full matching as a minimum-cost flow problem is established (Hansen and Klopfer, 2006); what the flow model contributes here is that every design in the package that a flow can represent compiles into it, so one solver, one certificate and one warm start serve all of them. Those are the six rows of the table: one-to-one, k-to-one, with replacement, blocked and exact, full matching, and fine and refined balance. The designs that sit outside it sit outside because their answer is not a flow: `cem_match()` coarsens and strata are read off directly, `subclass_match()` cuts the propensity score into subclasses, and the moment constraints of `cardinality_match()` are dualised and searched by branch and bound rather than represented in a network.

Table 1: How each matching design is stated in the internal flow model. The designs differ in the capacity bounds their arcs receive and in how many networks are built, rather than in which solver runs.

Design	Flow formulation
One-to-one matching	Unit-capacity bipartite flow
k-to-one matching	Capacitated bipartite flow, treated supply k
Matching with replacement	Control capacities above one
Blocked and exact matching	Independent networks per stratum
Full matching	Lower and upper capacitated circulation
Fine and refined balance	Category nodes with capacity bounds

Compiling a design, solving it, and checking the answer stay three separate steps, because the third is worth something only while it is separate: a solver status is not itself an independently checkable certificate. The flow and the potentials go into `verify_flow()` as inputs to a check, and the arcs the flow is checked against are stated independently of it.

```
prob <- list(n_nodes = 4, supply = c(2, 1, -2, -1),
            arcs = data.frame(tail = c(1, 1, 2, 2), head = c(3, 4, 3, 4),
                              lower = 0, upper = 2, cost = c(1, 3, 2, 1)))
cert <- verify_flow(c(2, 0, 0, 1), prob, potential = c(0, 0, 1, 1))
```



```
c(certified = cert$certified_optimal, gap = cert$duality_gap)
```

```
#> certified      gap
#>         1         0
```

Two solvers read that network. One is successive shortest paths with Johnson potentials (Johnson, 1977), reachable as `method = "csflow"`. The other, `method = "push_relabel"`, is cost-scaling push-relabel in the successive approximation of Goldberg and Tarjan (1990): a sequence of ϵ -optimal flows, each phase dividing ϵ , saturating the arcs the smaller ϵ no longer admits, and clearing the excess with two local operations. A push moves excess along an arc of negative reduced cost; a relabel lowers the price of a node holding excess and having no such arc, by exactly the amount that gives it one. Below $\epsilon = 1/(n+1)$ an ϵ -optimal flow on integer costs is optimal, which ends the scaling, so real costs are scaled to integers first.

Tolerance is where a flow certificate can overstate what it knows, and the scale of the potentials sets it. A design that ranks objectives lexicographically by weighting them, as `cardinality_match()` ranks cardinality above balance above distance, reaches potentials in the millions, where one unit in the last place is around 10^{-9} and an exactly optimal flow cannot meet an absolute tolerance of that size. The comparisons therefore scale with the arc's own magnitude, and the widest tolerance any comparison used comes back as `dual_tolerance`, so the verdict names the resolution it was reached at.

Node potentials therefore come back from every design rather than from the one-to-one case alone. Those potentials are what the next section prices against.

Certified implicit edge generation

A matching problem on n treated units and m controls has nm admissible pairs, and materialising them is what sets the size ceiling: at 50,000 against 100,000 the dense cost matrix alone is 40 GB. The lazy representation of the previous designs removes the matrix while leaving the search over all m columns intact: it traverses the complete graph an entry at a time rather than storing it. What `memory_mode = "implicit"` changes is the problem the solver ranges over. It optimises over a restricted pair set and certifies the result against the complete one, so the pairs outside that set are priced rather than searched.

The loop is column generation (Lübbecke and Desrosiers, 2005) over the pair variables of (1). A restricted master problem holds a subset of them, its optimal duals price the rest at $C_{ij} - u_i - v_j$, and the restricted solution solves the full problem as soon as no omitted variable prices negative. One thing separates it from the textbook setting: the omitted variables are enumerated rather than returned by an optimisation oracle, because a pair variable is an index rather than a combinatorial object to be searched for. Write $E_t \subseteq \{1, \dots, n\} \times \{1, \dots, m\}$ for the candidate set at round t and ϵ for the pricing threshold.

1. Seed E_0 with each row's w nearest admissible columns, w read off m as $6\lceil \log_2 m \rceil$ and capped at m .
2. Solve (1) restricted to E_t with the flow model of the previous section, warm-started from the incumbent flow and prices when $t > 0$. If it comes back short of a complete matching, repair the shortfall as below and repeat this step.
3. Read the master's potentials as assignment duals (u, v) .
4. Price the omitted pairs: $\bar{C}_{ij} = C_{ij} - u_i - v_j$ for admissible $(i, j) \notin E_t$.
5. Stop if $\bar{C}_{ij} \geq -\epsilon$ everywhere.
6. Otherwise set $E_{t+1} = E_t \cup V_t$, where V_t holds each row's most negative violators, and return to step 2.

The stopping rule is the certificate.

Proposition 1. *Let X solve (1) restricted to a pair set E with optimal duals (u, v) , and suppose $C_{ij} - u_i - v_j \geq 0$ for every admissible $(i, j) \notin E$. Then X is optimal for the complete problem.*

Proof. Optimality on E gives complementary slackness for X and (u, v) there, so every matched pair is tight and every unmatched column has $v_j = 0$, and the cost of X is $\sum_i u_i + \sum_j v_j$. The hypothesis extends dual feasibility from E to every admissible pair, so (u, v) is feasible for (2) on the complete problem and that sum is a lower bound on the cost of every assignment of it. X is one of them and attains the bound. $\square$

Nothing in the argument is numerical: the hypothesis is the sign of $\bar{C}_{ij}$ on the omitted pairs, and the conclusion is equality, not proximity. The loop prices against $-\epsilon$ instead of zero and the master is solved numerically, so the conclusion is weakened on both counts. Write δ for the duality gap that

master's own certificate reports, so that the restricted solution costs at most $\sum_i u_i + \sum_j v_j + \delta$. Under $\bar{C}_{ij} \geq -\epsilon$ the shifted potentials $u_i - \epsilon$ are dual feasible over every admissible pair, and shifting costs the dual objective $n\epsilon$, so the complete optimum is at least $\sum_i u_i + \sum_j v_j - n\epsilon$. The matching's excess over the complete optimum is therefore at most $\delta + n\epsilon$, to which a bounded error in evaluating the reduced costs of step 4 adds its own term. At $\epsilon = 10^{-9}$ and the largest shape in the benchmark below, $n\epsilon$ is 2×10^{-5} in the cost units, and every run reported there came back with $\delta = 0$.

The loop terminates because each pricing round adds at least one pair E_t did not hold: the round continues only when some omitted pair prices below $-\epsilon$, and that pair's row keeps it. So $|E_t|$ strictly increases, and it is bounded by nm . The worst case is that bound: a problem whose optimal duals price most of the grid in makes the loop build the complete graph and pay for the rounds that got there. `max_rounds`, 60 by default, is a guard rather than the termination argument; a run that reaches it returns `iteration_limit` rather than a certificate.

Each complete pricing sweep is $O(nm)$ arithmetic and stores nothing: the source is read one pair at a time and the reduced cost is consumed on the spot, so the memory the mode holds is the candidate set, nw pairs from the seed plus what the rounds add. Rows contribute at most `keep_per_row` violators each, five by default, so a round adds at most $5n$ pairs and never fewer than one. Selection is by strict improvement on a row's worst kept reduced cost over a scan running columns ascending, so ties keep the lower column index and the same input gives the same candidate set at every round.

A master that comes back short of a complete matching has no dual solution to price with, so the shortfall is repaired rather than priced. Hall's condition (Hall, 1935) names a row set S whose neighbourhood $N(S)$ in the restricted graph is too small to go round. Every restricted edge out of S already lands in $N(S)$, so only columns outside it can move the deficiency, and the re-seed reads the full source over the rows of S and takes columns from outside $N(S)$. When there are none, $N(S)$ is that neighbourhood in the complete problem too, and the result is infeasible with $(S, N(S))$ as its witness, naming the units short of partners rather than only the fact.

So the sparse solution is optimal for a problem that was never built, and `certify = TRUE` runs that check and returns it.

Two quantities in `$search` describe what that cost. `candidate_edges` against `possible_edges` is the graph that was built against the complete one that was not, and `edges_evaluated` counts distance evaluations over all rounds, so a pair priced three times counts three times. They move in opposite directions: the loop holds a small graph by evaluating distances more than once.

Round count is the quantity to control, since each round prices every omitted pair again, and it is what the seed rule of step 1 is sized for: a seed too narrow buys what it needs a round at a time, a seed too wide pays for columns the matching never uses. The rule clears the two-round floor, one seed and one sweep that finds nothing to add, on the problems below, and a run reports the width it used as `$search$seed_width`.

```
m_imp <- match_couples(treated, control, vars = covars, auto_scale = TRUE,
                      memory_mode = "implicit", certify = TRUE)
m_imp$certificate$certified_optimal

#> [1] TRUE

unlist(m_imp$search[c("seed_width", "n_rounds",
                     "candidate_edges", "possible_edges")])
```

```
#>      seed_width      n_rounds candidate_edges possible_edges
#>           54             2         10800         60000
```

The certificate the loop returns is the numerical one, which is why the ϵ version of Proposition 1 is the statement it carries. Dual feasibility over the pairs the master omits comes from the pricing bounds, which ask whether anything prices below $-\epsilon$ rather than evaluating each omitted pair, so the exact conclusion of the certification section is not reachable from a scan assembled that way, and the certificate reports that as its arithmetic.

Constraints tighten the loop rather than complicating it, since a caliper or a distance ceiling is checked before a distance is computed and removes pairs from the set to be priced.

Two neighbours are worth separating from this. Abeywickrama et al. (2021) also decline to build a complete cost matrix, deferring exact edge costs behind an inexpensive, application-specific lower bound, so what they save is arithmetic; the pricing here needs nothing of the cost function beyond the ability to evaluate a pair and saves storage instead, since every omitted pair is still evaluated once per round. The metric-tree pricer below is where the two meet. Graded matching (Yu and Rosenbaum,

2022) also enlarges a sparse network only as far as it has to, but a grade is fixed before the match, while the reduced cost that admits a pair here is a property of the solution the master has just returned, so the loop stops on a statement about the complete graph rather than on the grades running out.

The mode is available on `match_couples()` and `assignment()`, and it is requested rather than selected: `memory_mode = "auto"` does not choose it, because the worst case above is a real one. Its scope is the unit-capacity assignment. A full match's column nodes carry capacities above one, so a column dual there is not the assignment dual this pricer reads. A general flow arc does carry a reduced cost of its own, from the potentials at the nodes it joins, so pricing one is available in principle; it is this implementation that stops at the assignment. How small a graph the loop ends up holding, how many pricing rounds it takes and how it behaves on geometries built to be hostile to it are measured in the empirical evaluation.

Warm-started matching paths

A caliper is rarely known in advance. The usual practice is to sweep it, watch what the matched sample and its balance do, and choose a point, which means solving the same problem many times over. `match_path()` solves the sweep as one sequence.

```
p <- match_path(treated, control, vars = covars, auto_scale = TRUE,
               vary = "max_distance", values = c(1.0, 1.2, 1.5, 2.0, 3.0))
p$path[, c("max_distance", "n_matched", "total_distance", "certified")]

#> # A tibble: 5 x 4
#>   max_distance n_matched total_distance certified
#>   <dbl>      <int>      <dbl> <lgl>
#> 1         1         200        62.6 TRUE
#> 2         1.2        200        61.5 TRUE
#> 3         1.5        200        59.9 TRUE
#> 4         2         200        59.9 TRUE
#> 5         3         200        59.9 TRUE
```

The sweep shows where the constraint stops binding: the total distance falls until 1.5 and does not move above it, since a cut that wide forbids nothing the unconstrained optimum uses. Reading that off a path is the point of solving the sweep rather than one value.

Each point resumes from the matching the point before it found. Raising the distance cut admits pairs while leaving that matching feasible, so what remains is to repair reduced-cost violations on the arcs that just became admissible, which is one round of the pricing loop of the previous section; an independent call at that value pays for a solve from cold. One mechanism serves the sweep and the single solve. A value whose admissible pairs cannot support a complete matching is reported rather than approximated, coming back with no matching and the Hall witness beside it, so a sweep that starts below the feasibility threshold says where that threshold is. The ascending direction is a correctness requirement rather than a convention: a descending sweep withdraws pairs the current matching may be standing on, which breaks feasibility and needs a repair phase the loop does not have, so a descending sequence is refused with a message saying why.

The return value is a `couplr_path`, whose `$path` carries a row per point with the matched count, total distance, seconds, rounds and whether it was certified, and whose `$balance` carries a row per point per variable. `certify = TRUE` is the default here, and a point's certificate is what says its matching is the optimal one at that value rather than the one the warm start happened to reach. What a sweep costs against solving each point independently is measured in the empirical evaluation.

Memory representations

The dense representation of the cost matrix is the binding constraint on problem size long before runtime is. A 100,000 by 100,000 problem needs 80 GB as doubles; the same units described by 100 covariates need 80 MB. `couplr` exposes this as `memory_mode`, with values `"dense"`, `"lazy"`, `"implicit"` and `"auto"`, differing in what the solver ranges over. The dense path holds the complete matrix; the lazy path holds none and still considers every pair, evaluating each distance when it is asked for, over Jonker-Volgenant and auction with the built-in metrics; the implicit path runs the edge-generation loop described above. All three return the same assignment.

Under `"auto"` the package decides. Below ten million cells, about 80 MB dense, it skips the check, since probing free memory on every ordinary problem would be pure overhead. Above that

it estimates what a dense solve peaks at rather than what the matrix occupies, at 10 times the raw cell bytes, and compares that against available system memory read per platform. The Performance section reports the peaks that multiplier sits above. If the estimate exceeds half of what is available, "auto" switches to the lazy path where one exists and warns where one does not, naming blocking, greedy matching or a smaller problem. Detection failure is treated as unknown rather than as success: a fixed 4 GB threshold takes over, so the guard never silently disappears on an unrecognised platform.

5 Implementation and verification

The package is organised in three layers. The first turns user input into a cost source: it validates the data frames, screens covariates, applies scaling and weights, and settles how distances are obtained and which pairs are admissible. The second compiles the requested design into the flow model and solves it, which for the plain one-to-one case is (1). The third builds result objects and their methods. The object model, the C++ organisation of the nineteen solvers and the dependency footprint are in the supplementary material.

Automatic solver selection

method = "auto" inspects the problem and picks a solver, and the rules are few.

1. Problems with at most eight rows and eight columns go to exhaustive enumeration, which is exact and faster than setting up a general solver.
2. Cost matrices whose finite entries are constant or lie in $\{0, 1\}$ go to the Hopcroft-Karp style routine, which exploits the absence of a real cost scale.
3. Everything else goes to Jonker-Volgenant.

Sparsity and aspect ratio carried rules of their own, sending a mostly forbidden matrix to lapmod and a wide one to shortest augmenting paths. Both were slower than the default in the regime they were written for, so both were removed; Table 4 is the measurement, and both solvers stay reachable by name.

Rule 2 reads the entries, and so does the rejection of NaN inputs that runs for every method, so choosing a solver costs one C++ pass over the cost matrix, returning every quantity the rules need without allocating and without a second scan. That leaves the inspection a linear read in front of a superlinear solve, so its share of the runtime falls as problems grow, and the dashed line in Figure 1 sits on the solver the dispatcher selects. The pass is skippable for callers who already know the shape of their problem, by naming the method.

Independent verification

Certificates are the check that survives leaving the test suite. `verify_assignment()` and `verify_flow()` run in $O(nm)$ and $O(\text{arcs})$ respectively without solving anything, so a user can run them on their own problem, in exact arithmetic or at their own tolerance, against a solution the package produced.

Testing and reproducibility

A smoke test establishes that a function returns an object of the right shape; for a solver the standard is whether the assignment is the optimal one. Every optimal solver is checked against exhaustive enumeration on randomised instances covering integer and fractional costs, square and rectangular shapes, minimisation and maximisation, and forbidden edges. The reference is the same brute-force solver the dispatcher uses below nine units, so the verified and the shipped path are one implementation. Constrained matching is checked against an enumeration over all partial matchings of a small instance, which supplies both halves of its lexicographic objective. The paths that avoid materialising the problem are checked against the path that materialises it: the implicit loop and the dense solve must return the same total and the same pairing wherever the dense solve runs at all, and a warm-started point must return what an independent solve at that value returns. Two implementations that should agree exactly are the useful kind of oracle, since a disagreement is a defect rather than a matter of degree, and the next section reports both comparisons on the sizes it covers.

The package carries 138 R test files and 37 C++ test files, covering the R layer, the solver bodies compiled outside R, the C++ wrappers, dispatch rules and the matching designs.

Every number the next section reports comes from a script that ships with the article. Each caches its CSV under `paper/` and re-measures only when that file is moved aside; `paper/run_bench_suite.sh`

runs them in order on one machine, which is how these numbers were produced, and the scaling, implicit, implicit-grid, regime and memory scripts take --quick for a reduced grid. The environment block beside the CSVs records the machine, the R version, the package version and the commit.

6 Empirical evaluation

Edge generation, robustness and memory

Table 2 reports the three memory modes on those same problems. The dense and lazy arms are pinned to Jonker-Volgenant, so they differ only in representation; the implicit arm carries the flow model as its restricted master by construction, which is a difference between the arms and is the definition of the mode rather than a choice.

Table 2: One-to-one optimal Mahalanobis matching by memory mode, wall-clock seconds on a single core of an Apple M4 Pro. Graph built is the arcs the implicit loop ended up holding as a share of the complete problem's. Distances is its evaluations over all rounds as a multiple of the complete pair count, so a pair priced twice counts twice. All three arms were timed in one session; the dense arm was run at the four sizes where its pairing can be compared against the loop's, and the scaling table below carries dense timings at the two largest sizes. Every cell returned the same total distance.

Problem size	dense	lazy	implicit	Graph built	Distances	Rounds
167 + 333	22 ms	8 ms	12 ms	16.22%	2.00	2
667 + 1,333	161 ms	53 ms	56 ms	4.95%	2.00	2
1,667 + 3,333	837 ms	351 ms	299 ms	2.16%	1.96	2
3,333 + 6,667	3.14 s	1.59 s	1.09 s	1.17%	1.88	2
6,667 + 13,333	not run	6.38 s	3.9 s	0.63%	1.71	2
16,667 + 33,333	not run	62 s	20.2 s	0.29%	1.42	2

Three things in that table go together. The loop holds a small graph, 16.22% of the pairs at the smallest size and 0.29% at the largest, a share that falls as the problem grows since the pairs a matching can use grow more slowly than the pairs a problem has. It pays for that in arithmetic, at 1.42 to 2.00 times the complete pair count in distance evaluations, so at eight covariates it computes more distances than one full sweep would. And it is still the fastest arm at five of the six sizes, because what it avoids is not the arithmetic but the graph. At 16,667 + 33,333 it finishes in 20 seconds against 62 for the lazy path. At 667 + 1,333 the lazy path is faster, at 53 ms against 56 ms, which is where the loop's fixed costs are still visible against a problem this size.

Every run in the table settled in 2 rounds, one seed and one sweep that found nothing to add and certified. Every one came back certified with a duality gap of zero. On the four sizes where the dense solve can also be run, the two agree to within $3.4\text{e-}13$, and the pairing is identical unit by unit, so the loop returns the assignment rather than an assignment of the same cost.

The implicit column is a representation and a solver at once, so comparing it with the dense column reads both differences together. At 3,333 + 6,667 the materialised matrix takes 3.14 seconds under Jonker-Volgenant and 4.17 under the flow solver the restricted master belongs to, against 1.09 for the loop, so what the loop saves is not what a change of solver would give. The supplementary material carries the decomposition.

One cloud is not a characterisation, and the table above reports one. The supplementary material sweeps the loop around that cloud one factor at a time: covariate counts from 2 to 32, seven point clouds including clustered points, a displaced treated group, coordinates on a coarse lattice so that distances tie, one contested core where every treated unit's nearest controls are the same few, and a shell where every treated-to-control distance is nearly equal, three caliper settings including the tightest value each problem is still feasible under, and five sizes, each at several seeds. Across the 20 cells that grid holds, the loop settled in 2 to 10 rounds and ended holding between 0.29 and 73.32 percent of the complete pair count, at 0.24 to 6.67 times that count in distance evaluations. All 20 cells certified, at a largest absolute duality gap of $3.64\text{e-}12$, and 18 of the 18 cells small enough to run the dense path beside the loop returned the same pairing unit by unit. The margin over the lazy path is where the clouds separate: it runs from 11.13x on shifted to 1.09x on heavy_tailed.

Two rounds is what an ordinary cloud costs and it is not what the loop is bounded by, so the grid's hostile cell is worth reading on its own. In the contested core, where every treated unit's nearest controls are the same few, the duals travel a long way before they stop pricing pairs in. At 10,000 units it takes 9 to 10 rounds, ends holding 73 percent of the complete pair count rather than a fraction of one

percent, and evaluates 6.7 times that count in distances. It still certifies, and it is still 3.2 times faster than the lazy path on this cloud. What it shows is the worst case of the previous section happening: the graph the loop ends up holding is most of the one it set out to avoid, and it has paid for the rounds that built it. This is why the mode is requested rather than selected.

The claim that memory rather than time is what bounds a dense matching has so far been made with edge shares, which are a proxy for it. Peak resident set size is the quantity itself, and it is a property of a process rather than of an expression, so each arm here is a fresh R session that loads what it needs, runs one matching and exits, measured from outside. A session that loaded the same packages and matched nothing is the floor each arm is read against.

At $6,667 + 13,333$ the dense path peaks 6234 MB above that floor, the lazy path 22 MB and the loop 128 MB, against 27428 MB for `optmatch` and 8640 MB for `MatchIt`. Two readings follow. The loop is not the smallest of the three: the lazy path holds 22 MB against the loop's 128 MB, since it stores no graph at all while the loop stores the candidate set it has built. What the loop buys over it here is time, and the ceiling both of them remove is the dense path's. The second reading is the guard's. Building the matrix and stopping there accounts for 1095 MB on a matrix whose entries occupy 711 MB, so holding it costs 1.5 times its cells and solving over it costs 8.8 times them. That gap is preparation copies and the solver's own workspace, and it is what the automatic mode has to read: a guard sized on the matrix would let a dense solve start on a machine it does not fit. `estimate_dense_matrix_mb()` estimates the matrix at 4 times the raw cell bytes and `estimate_dense_solve_mb()` the peak of the solve at 10 times them, and `memory_mode = "auto"` compares the second against available memory. Neither multiplier is fitted to one cell. Across 5,000, 10,000 and 20,000 units the solve peaked at 9.5, 7.2 and 8.8 times the cell bytes and the matrix at 1.8, 1.7 and 1.5 times them, so each default sits above the largest of those rather than at any one of them. That is the direction the guard has to err in, since it exists to refuse a solve that will not fit rather than to predict where the peak lands. Every arm and size is in the supplementary material.

Warm-started paths

Table 3 reports a sweep of 20 caliper values solved as one path against the same 20 values solved independently. The grid is the data's own: its tight end is the tightest caliper the problem still admits a complete matching under, found by bisection, and its wide end is the median treated-to-control distance, which admits half the pairs and constrains nothing.

The table leads with the wall clock for the whole sweep, which is the number a user meets: it carries the feasibility check, the balance reading each point takes, the conversions in and out of the solver and the rest of the R-level work. Both sides pay that per point, since an independent solve here is a path of one value rather than a separate entry point, so the comparison is like for like. The solver's own time sits beside it as a diagnostic, saying how much of the saving is the warm start rather than the sweep's shared setup.

Table 3: A caliper sweep solved as one warm-started path against the same values solved independently, end-to-end wall clock for the whole sweep on a single core of an Apple M4 Pro. Solve time is the same comparison on the seconds the solver reports for itself, summed over the points, which excludes the R-level work both sides do. Rounds is the pricing rounds each side took, summed over the sweep.

Problem size	As a path	Independently	Ratio	Solve time	Rounds
167 + 333	0.057 s	0.136 s	2.39x	2.88x	22 / 40
667 + 1,333	0.489 s	1.04 s	2.12x	2.15x	23 / 40
1,667 + 3,333	2.88 s	6.12 s	2.13x	2.13x	21 / 40
6,667 + 13,333	40.9 s	84.6 s	2.07x	2.07x	21 / 40

The saving comes from the rounds column. A warm-started point starts from a matching that is still feasible at the new value, so what it has to repair are the reduced-cost violations on the arcs the wider cut just admitted, which the loop reaches in about one round per point. A point solved independently starts from a seed and pays for the seeding sweep as well.

Correctness is the part of this comparison that has to hold before the timing means anything. Every point on the path returns the status and the matched count its independent solve returns, and the totals agree to 0 relative, so the path is the same sequence of matchings that solving each value from cold would produce.

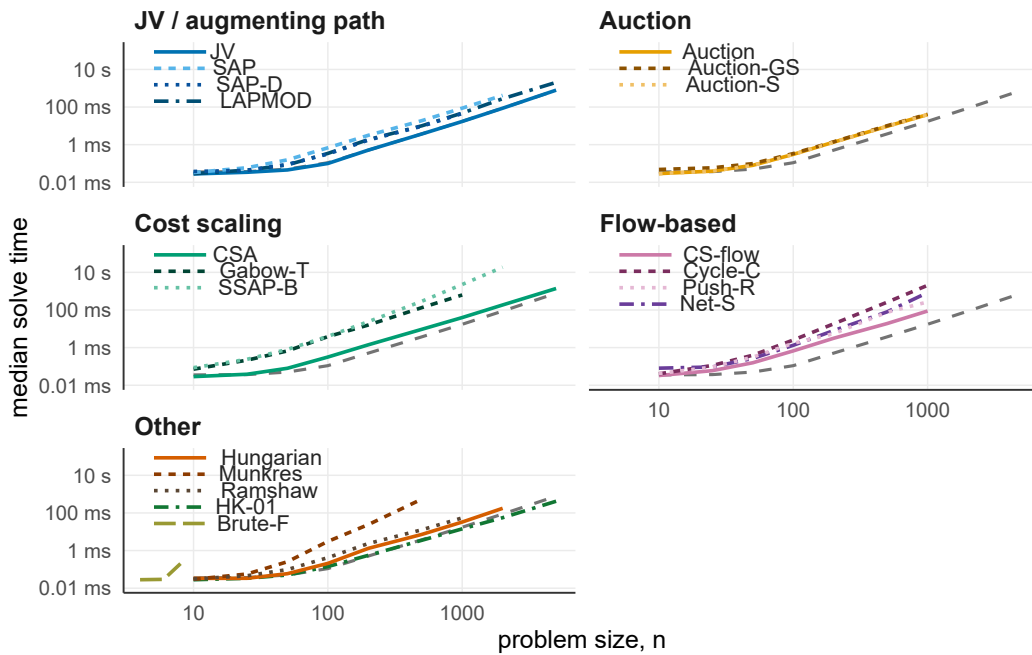


Figure 1: Median wall-clock solve time against problem size for the nineteen assignment solvers in couplr, grouped by algorithm family on shared log-log axes. The two special-purpose solvers in the Other panel run on their own inputs and are not comparable to the rest: HK-01 is timed on binary cost matrices, and Brute-F only up to $n = 8$. The dashed grey line repeated in every panel is the automatic dispatcher on the uniform integer costs.

Solver portfolio and automatic dispatch

Figure 1 reports median wall-clock solve time against problem size for all nineteen solvers, grouped by family. We draw costs as integers uniformly from $[1, 10,000]$ on square matrices and report medians of five replicates on a single core of an Apple M4 Pro. The slower families are capped at smaller sizes so that the scalable solvers can be measured to $n = 5000$ within a reasonable budget. All timings in this section were measured on 2026-08-30 under R 4.5.3 with couplr 1.7.0 at commit a63b760, [optmatch](#) 0.10.8 and [MatchIt](#) 4.7.2.

Jonker-Volgenant with its warm start is the fastest general-purpose solver at every measured size, which is why it is the dispatcher’s default. The auction and cost-scaling families are competitive at small sizes and fall behind on dense uniform costs, where their advantage, tolerance of degenerate structure and a better asymptotic bound for Gabow-Tarjan, does not pay. General flow formulations are the slowest, as expected of machinery solving a larger problem than the one posed. The two special-purpose routines run on the inputs they exist for, exhaustive enumeration below nine units and the Hopcroft-Karp style solver on binary costs, so their timings are not comparable with the rest and are shown only to establish that each is practical on its own input. These orderings hold for dense uniform integer costs on square matrices, which is the whole of what the figure establishes, and they fix the dispatcher’s default.

The grid behind Table 4 is what reduced the rules to three. It crosses the properties the solver figure holds fixed: whether the finite costs have a scale worth exploiting, how much of the matrix is forbidden and whether what remains is one component, and how far from square the problem is. Eight cost regimes, from uniform integers through binary, constant, heavily tied and heavy-tailed costs to euclidean distances over uniform and clustered points, are crossed with six admissibility patterns, complete, unstructured at four densities from 60 percent of the entries finite down to 1 percent, and four disconnected components, and with three aspect ratios from square to one row per ten columns. That crossing is the base tier. The large tier repeats a subset of it at three times the size rather than the whole: the regimes and patterns where a rule that holds at 500 rows might fail at 1500, against the solvers that reach that size inside the budget. The table therefore covers 189 cells, 144 of them the complete base crossing and 45 the large-tier subset. Each cell is generated as several independent instances, each instance is timed several times, and every solver in the panel sees the same instance. The solver “auto” selects is timed as a solver of its own, so its number carries the probing pass the rules cost, and it is read against the fastest named solver in the same cell.

Over the 189 cells the dispatcher picked the fastest solver of the panel in 94 of them, at a median

Table 4: Each dispatch rule over the cells of the regime grid where it fired. Solver is what the rule selects. Fastest counts the cells where that solver was also the quickest of the panel. Median and worst are the time the dispatched solver took over the time the cell’s fastest named solver took, so 1.00x is a rule that picked the best available solver. The large tier times a reduced panel that omits the special-purpose solvers, so a cell whose dispatched solver is not in it reads below 1.00x.

Rule	Solver	Cells	Fastest	Median	Worst
no earlier rule applies	jv	144	59	1.04x	9.95x
no cost scale to exploit	hk01	45	35	1.00x	1.93x

1.02x of the cell’s best time. A cell reads below 1.00x for one of two reasons. The large tier times a reduced panel, so on its 9 binary cells the solver the dispatcher named was not among those it is read against; on the 36 cells of that rule whose panel does hold its solver the median is 1.02x. Elsewhere the two sides are separate timings of one solver and scatter by a few percent either way. The two rules that were removed can be read off the same table. Over the 96 cells with at least three columns per row, where the rectangular rule used to fire, the default takes a median 1.02x; over the 96 cells with more than half the entries forbidden, 1.14x. Neither is a regime the default needs help in. Where it is furthest behind, at 9.95x, the winner is auction_scaled on a 500 x 500 problem with heavily tied costs, which is a structure no rule in the table reads and none was added for: fitting one to this grid would score it on the measurements that chose it.

The full grid, every solver in every cell, is in the supplementary material.

Comparison with established alternatives

Balance

Speed only matters once the packages agree on the answer. We fix the task in Figure 2, one-to-one optimal matching on a Mahalanobis distance with pooled within-group covariance, and run it on the LaLonde NSW data (LaLonde, 1986; Dehejia and Wahba, 1999), 185 treated units, 429 controls and eight covariates. The pooled within-group convention is the one `optmatch::match_on()` uses by default and the one `couplr` adopts.

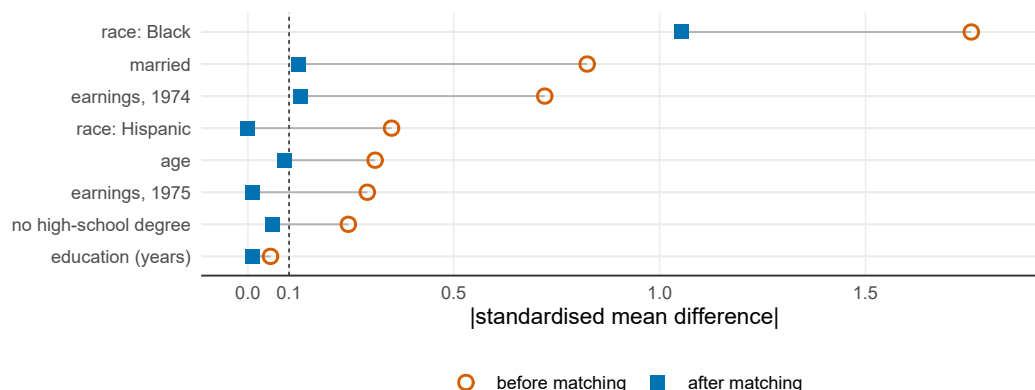


Figure 2: Absolute standardised mean differences on the eight LaLonde NSW covariates, before and after one-to-one optimal Mahalanobis matching with pooled within-group covariance. `couplr`, `MatchIt` and `optmatch` agree to three decimal places after matching, so a single matched point is shown per covariate. The dashed line marks the conventional 0.1 threshold. The indicator for Black respondents starts far outside the plotted range at 1.757 and remains at 1.053 after matching, a residual imbalance that no one-to-one matcher can resolve on this data.

The three packages produce identical absolute standardised mean differences to three decimal places, so one matched point is plotted. Marginals could still hide different assignments, so we score all three on one objective: the same Mahalanobis distance matrix, summed over each package’s chosen pairs. Each matches all 185 treated units, at a total distance of 304.042 for `couplr` against 304.043 for both alternatives, a relative difference of 3×10^{-6} . That is agreement on the objective to the precision reported here rather than a demonstration that all three attain the same minimum, and separating those two readings is what a certificate is for: `couplr`’s matching comes back certified optimal for this cost matrix, and the same check runs on either alternative’s pairing against the same duals.

Five of the eight covariates fall below 0.1 after matching, married and 1974 earnings sit just above at 0.124 and 0.128, and the indicator for Black respondents drops from 1.757 to 1.053 without approaching it. That is a property of this dataset rather than of any package: the control pool does not hold enough Black respondents for a one-to-one match to balance that indicator.

Scaling

Table 5 reports wall-clock time on synthetic problems with the same eight-covariate structure, at six sizes with a one-to-two ratio of treated to control units, so every problem here is rectangular. Every package in this comparison materialises the distance matrix, so we time `couplr` with `memory_mode = "dense"`. Each size is generated as several independent instances with the timing repetitions inside them, and the table reports the median and the interquartile range across instances: repetitions of one matrix measure the clock, and what varies between problems is what a reader wants to know.

Table 5: One-to-one optimal Mahalanobis matching, wall-clock time by problem size. Treated to control ratio 1:2, eight covariates, pooled within-group covariance, single core with single-threaded BLAS on an Apple M4 Pro. Each cell is the median over independently generated instances of that size, with the interquartile range across instances in brackets and the timing repetitions taken inside each instance. The 50,000 row carries one instance and is a single run rather than a median, so no range is reported for it. The `optmatch` option `max.problem.size` was set to `Inf` for the two largest sizes. `int overflow` marks an integer-size overflow inside the `optmatch` back end reached through `MatchIt`; `timeout` marks a run exceeding the 600-second cap.

Problem size	<code>couplr</code>	<code>optmatch</code>	<code>MatchIt</code>
167 + 333	17 ms (17-19)	120 ms (111-121)	134 ms (132-139)
667 + 1,333	140 ms (139-143)	1.75 s (1.67-1.77)	2.14 s (2.12-2.16)
1,667 + 3,333	789 ms (779-821)	12.8 s (12.7-12.8)	16.3 s (16.2-16.3)
3,333 + 6,667	3.16 s (3.1-3.22)	60.5 s (59.8-60.5)	74.2 s (73.5-74.3)
6,667 + 13,333	11.6 s (11.2-12.1)	287 s (284-290)	int overflow
16,667 + 33,333	76.3 s	timeout	timeout

The margin widens with problem size. `couplr` is about 7 times faster than `optmatch` at $n = 500$ and 25 times faster at $n = 20,000$, where it finishes in 11.6 seconds against 4.8 minutes. `MatchIt` runs 8 to 23 times slower up to $n = 10,000$, which is the largest size it completes before `matchit(method = "optimal")` aborts inside the `optmatch` back end with an integer-size overflow. At $n = 50,000$ `couplr` finishes in 76 seconds and both alternatives exceed the 600-second cap.

What the table compares is three workflows on one task, from data frame to matched pairs, not three implementations of one algorithm. The packages do not state the same problem: `optmatch` solves a minimum-cost flow problem, which the full and variable-ratio designs it is built around need and a plain one-to-one problem does not, and `MatchIt` reaches that same back end, while `couplr`'s dispatcher sends this problem to Jonker-Volgenant. Reading the margin as a faster inner loop would need a benchmark holding the formulation fixed across packages, which this one does not.

The lazy path removes the memory ceiling as well. The same $n = 50,000$ match under `memory_mode = "lazy"` completes in 62 seconds without ever building the distance matrix, returning the assignment the dense path returns. At $n = 20,000$ it takes 6.4 seconds against 11.6 for the dense path, so on these problems recomputing distances is cheaper than materialising and traversing the matrix.

Feature coverage

Table 6 lists the features that separate the three packages. Rows where all three offer the feature, covariate distance, propensity-score matching, exact blocking and a cost-matrix entry point, are omitted.

Table 6: Selected assignment-layer features, omitting areas where the alternatives lead, among them the breadth of designs reachable through a single MatchIt call and the maturity of optmatch’s full matching. optmatch’s calipers threshold one distance object at a time, so a per-variable set requires combining objects, and both alternatives reach the solver as a padded or variable-ratio formulation rather than as a rectangular cost matrix. The certificate row records whether a package exports a function returning duals or verifying optimality, which neither does: optmatch attaches its flow solver’s node prices to a fullmatch() result in an undocumented MCFsolutions attribute, and the function evaluating the primal against them is internal. That row was assessed on 2026-08-26 and the others on 2026-08-09, both against MatchIt 4.7.2 and optmatch 0.10.8.

Feature	couplr	MatchIt	optmatch
Per-variable calipers from a named vector	yes	yes	partial
Rectangular problems without padding	yes	via padding	via padding
Sparse cost support	yes	no	yes
k-best assignments	yes	no	no
Bottleneck (minimax) assignment	yes	no	no
Rosenbaum sensitivity bounds	yes	no	no
Dual potentials and optimality certificate	yes	no	no
User-selectable assignment algorithm	19 solvers	via optmatch	5 algorithms

7 Summary and discussion

We built couplr on the observation that one optimisation problem sits underneath a set of tasks R currently treats as unrelated. Making the assignment problem the organising object, rather than a hidden step inside a matching function, is what lets one package serve a causal-inference user starting from covariates and an operations-research user starting from a cost matrix. Three design consequences follow. Every flow-representable design compiles into one flow model, so node potentials come back from all of them. Those potentials make a solution independently checkable, which turns a reported status into a certificate. And a certificate is what makes it safe to solve a problem that was never built, since the loop that grows a sparse graph needs a statement about the complete one to know when to stop.

What that architecture buys is measurable. At $16,667 + 33,333$ the restricted graph settled at 0.29% of the complete pair count, returned a certified optimum, and ran 3.1 times faster than traversing the complete graph lazily. A sweep of 20 caliper values solved as one warm-started path cost a median 0.47 of solving each point cold, and across the 189 cells of the solver-regime grid the dispatcher ran at a median 1.02x of the cell’s fastest measured solver.

couplr has five limitations:

- The edge-generation loop’s scope is the unit-capacity assignment: a full match’s column nodes carry capacities above one, so its column duals are not the assignment duals the pricer reads, and full_match() has no implicit path. That is a limit of this implementation and not of the method, since a general flow arc has a reduced cost from the potentials at its ends. Its pricing is a block scan over the complete pair set, so at eight covariates the loop evaluates more distances than a single full sweep and wins on what it does not build rather than on what it does not compute.
- The lazy cost path covers Jonker-Volgenant and auction with the built-in metrics; a user-supplied distance function still requires a dense matrix.
- cardinality_match() answers a moment-constrained problem by branch and bound under a node and time limit, so such a run can stop with a gap rather than a certificate and reports which it reached; fine and refined balance, which the network represents directly, is answered certified.
- The dispatcher reads two properties and otherwise defaults to Jonker-Volgenant, which the regime grid supports across the structures it covers. It is not the best choice everywhere on that grid: scaled auction is three to four times quicker on the small square cells with most of the matrix forbidden, and the dispatcher leaves that on the table rather than adding a rule fitted to the grid that would score it. Naming a solver is what a caller with such a problem should do, which is why every solver stays reachable by name.
- Matching on observed covariates cannot address unmeasured confounding; the sensitivity analysis quantifies exposure to it rather than removing it.

One extension follows from those limitations. A metric-tree pricer would attack the arithmetic the block scan leaves in place. Discarding a subtree S of controls for row i has to be decided without visiting

it, so the test is $L(i, S) - u_i - \max_{j \in S} v_j \geq 0$, where $L(i, S)$ is a lower bound on $\min_{j \in S} C_{ij}$ computed from the node's summary rather than the minimum itself. Mahalanobis distance is Euclidean after whitening, so one tree serves both metrics, but each node then has to carry two summaries: a centre and radius in whitened coordinates, which is what $L(i, S)$ reads, and an axis-aligned bounding box in the original covariates, since a caliper is a box there and a whitened ball cannot exclude one. Calipers are the case for the second summary, being where matching users actually work. Where such a bound prunes, the prune certifies that its subtree holds no violating edge, and the prunes together with the leaves the scan examines certify the pricing result.

couplr is available on CRAN and developed at <https://github.com/gcol33/couplr>, with documentation at <https://gillescolling.com/couplr/>. Bug reports and feature requests go through the GitHub issue tracker. No external funding supported this work.

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